

John Boik: "CogNarr Ecosystem: Facilitating Group Cognition at Scale"

29:56

welcome John Bo PhD to the fourth active inference Symposium we're really pleased to have you here and looking forward to your session cogar ecosystem facilitating group cognition at scale and over to you all right thanks

Ian um again my name is John Bo I'm a a fellow at the Active Institute active inference Institute

um and I will be talking about a cogar ecosystem to to facilitate cognition at scale as Ian said uh I do want to say that uh the links for this there's a bunch of links on The Institute project page uh I think I think these slides will eventually be made public but in the meantime if you're interested you can just simply search on cogar and Kota or cogar and my last name bo and you'll find The Institute project page there's lots of links

there all right so is the cogar ecosystem uh cogar means cognitive narrative and in concept it's uh it's an open- Source online extensible ecosystem uh Technologies tools libraries to facilitate group cognition uh especially for large groups and although there are many many use cases uh uh the the use case that's that one of the Prime use cases is uh deliberation strategizing Collective problem solving and decision making in the large group

setting cogar essentially helps what I like to say is it helps users to tell

their story that is convey their beliefs
uh to a group about how the world Works
relative to some situation and even more
specifically convey their belief model
to a group make make the belief model
explicit and transparent to the
group um so they might uh they might
answer or address questions like what's
the problem or situation what happened
how did it happen What might happen next
why did it happen who does it affect why
who do it involve what does it mean what
should be done what can be learned all
of these kinds of
questions um cogar allows PE allows
users to tell a rich story and by Rich I
mean potentially long complex nuanced
Dynamic conditional uh uh
um quite different quite in comparison
quite different from some kind of online
collaborative architecture that for
example asks you to a yes or no question
or ask you to give a thumbs up or thumbs
down on some
issue uh now this is uh this is this
cogar will allow a user to convey very
rich information and there's a back-end
computational system that assists the
digestion of that information for the
for so that the group can better
understand the the many different
stories so the computational system
assists and guides users in construction
stories uh helps the group digests the
information make sense of it
um uh assesses stories for their quality
you know is the is the story on topic is
it

comprehensible that kind of thing and then the backend system also performs inference for queries that uh that various group members might have about a story or for a Storyteller it is important to say that the system the computational system does not predict the future that is it's not going if a bunch of people if the issue at hand say is garbage collection in a local community this the uh computational system is not going to assess stories and then predict which of them is most realistic that's not really its purpose what it what its purpose is is to convey to make transparent uh and explicit the belief of members and uh the members when they create their story when they tell their story they include what they think will happen in the future the predict they make pred the users make predictions um and also the group assesses the realism then of Any of of the stories that are submitted so what is a cogar group uh a group is is a set of people that sense their world process information anticipate comes evaluate options make decisions and in other words we can think of a group as a as a as an organism that performs cognition um and then the question is how what what design of cogar might actually facilitate functional cognition as opposed to dysfunctional cognition so examples of groups could be you know

every a group informal group who like
polka dot shirts uh a social club groups
Civic organizations professional groups
corporations nonprofits political
entities such as cities and
more uh so groups can range from formal
to informal flexible to structured
temporary to permanent small to large
autocratic to democratic the the the
platform is really a tool that can be
used by for many use cases by many kinds
of groups for many
purposes there are of course some rules
to keep it all functional and
transparent
and when I say large group what does
that mean uh uh often in the literature
when they talk about a lar large group
uh uh deliberation and you know
cognition they might be thinking of
dozens of people but uh here we're
thinking about perhaps dozens hundreds
thousands millions there this should
scale up to you know cities states and
perhaps even a nation who knows you know
if if I should say if cogar were ever
built if it works has advertised uh you
know this is all possible and and maybe
I should also say at the moment this is
just there's I published uh two or i'
I've posted two story two articles on
cogar so far two research articles and
um that's all that's been done so this
is this is at the beginning this is the
a vision really of what this could
be I'm I'm presenting a
vision
um so the question is what kind of

design for cogar would allow some large group say a 100,000 people to to to um gather online synchronously or asynchronously and share the potentially complex Dynamic beliefs belief models about some situation how could that be done that's the question here and I obviously I have have some ideas on how that could be done so uh but first maybe I should say that that the cogar project is inspired by active inference and in particular active inference Pro from you know from my uh from the vantage point of this project active inference provides a description of functional cognition that is as I think everyone probably knows by now watching watching the seminar that uh that cognition we can idealize it as a maybe a four-part cyclic process of predict act sense and learn and and it involves um many components for example memory recall attention um and this the entire picture of that process can be um you know that as it plays out in a human or an agent it could be either functional or dysfunctional for example uh an an agent that that that cannot predict the future with accuracy is going to make decisions that are suboptimal in gen uh you know typically so that would be a a an example of dysfunctional cognition if you can't remember what happened or you can't uh predict accurate accurately what's going to happen in the future

those would all be IND indicative of
dysfunctional

condition so an act of inference an agent
favors choices that ensure the greatest
resolution of uncertainty under the
constraint as preferred outcomes are
realized and all of that provides uh you
know a framework such that um that the
components of cognition could be
recognized can each component can be um
focused on as as a design element and
the and the and the and the process of
cognition in any group could be assessed
whether whether that is a functional or
dysfunctional or somewhere in between
and of course that would be a you know
multivalued

scale so what is cogn what is cogar
really um cogar essentially is designed
to be part of the cognitive architecture
of the group that is using it that any
organism that performs cognition has
some kind of cognitive architecture for
for a for animal that you know involves
a nerv nervous system it involves you
know can for a human it can involve
external objects such as Computers and
Tools and um for a group it includes a
set of norms rules mechanisms and
institutions sensors analytic processes
systems and any other procedures that
facilitate or guide cognition and then
Co cogar is obviously designed to be a
component of the cognitive architecture
of the group that uses it and viewed
differently and I'll talk about this
more in a moment cogar technology
provides a means to digitize and make

explicit a person or an agent's belief model and in that sense the the some of the technology behind cogar could be used far beyond cogar uh could be used in medicine psychology and you know unlimited number of other applications of of how you would take you know an agent or a person's belief model Make That explicit in a way that computation can occur and um and you know include that into some kind of inference models or generative models so cogar stems from some past work that I've done and it's this past work the cogar and this past work are related in that all of it is about cognitive architecture in the past work I was looking at Social systems as cognitive architectures and by social systems I mean economic systems Financial systems governance systems I view those core societal systems as the as a means as a as a part of the cognitive archit architecture of the society that is using and that is we we think through our um governance systems we think through our economic systems we make decisions that that affect our world through our economic system for example and um the previous projects asked this question out of all conceivable societal systems designs and I don't mean conceivable in you know capitalism and socialism and the way it's been implemented in the past what I mean is like Blank Slate out of all

conceivable designs for uh economic
Financial governance systems which ones
but might be most fit for purpose and
that purpose is cognition in the active
inference set so reduction of
uncertainty accurate prediction of the
future uh attaining those conditions
that that allow for thriving of the
population uh those
things and I propose in this past work
uh a path really of how such a project
could be implemented how you know what
it would do how it would go how how
would you test new systems um
um and so on so I referred to this
earlier work and again the links are all
on the cogar project page if you can
find
that all right so what is uh the maybe
the core element of cogar is um what I
call a story graph and essentially it's
a meaning representation it's the it's
the explicit representation of a user's
belief model and this of course is a
very difficult problem to solve how do
you how do you take some belief model
that is in a is in say the mind of of a
person and put that down on paper so to
speak such that computation can occur
with it inference can occur with it this
a very very tough problem but I think
there's there's possible ways to to move
forward with it um I think I mentioned
or I should mention that there was two
papers published on cogar first paper is
more of an introduction overview the
second paper is really all about it
focuses on the potential meaning

representations um so story graph is the sort of the core Innovation on which kogar is based and it's story graphs are used to curate understand reason with you know conduct inference on high quality information about human experiences beliefs and expectations and essentially the way it works is um a user might type in or say um some passage a few sentences perhaps a thought and uh that Passage would be translated into a what I call a story graph fragment just a small little graph of meeting graph and that little fragment would then the system would merge that fragment into the growing story graph the overall story graph and then eventually by by uh by um inputting passages after passages after passages the story graph gets built up the meaning representation gets built up and then once it's complete or once it's almost complete then it can be translated into alternative meaning representations that might be appropriate for various you know logical models or other kinds of models probabilistic models um can maybe be translated more directly into some types of probabilistic models and could also be translated into a natural language so in in Time Event you know if kogar is successful over time it should be possible to have the system read back your story like tell me tell me you know read this back back to me and let me make sure that it's what I think what I'm intending it to be uh even you know

in the far future it might be possible
to make animations as an output of this
you know this a little movie of this is
what I think happen and this is what I
think will happen in the
future um so what is a meaning
representation I have a picture here of
a um of a abstract meaning
representation that's a very common one
used in the in the in the research world
the the the in in natural language
inference there's u in the research
world there's lots of different types of
of meaning representations this is just
one of them I'm using as an
example and this is for a story excuse
me a sentence a similar a similar
technique is almost always impossible to
apply to other crops such as cotton
soybeans and rice and you can just you I
won't go into detail about this but you
can kind of see that there's a graph and
various words are connected uh to to
each other ideas are connected to each
other concepts are connected to each
other and this kind of meaning
representation is used throughout
research and the story graph you can
think of a story graph as a much more
complex uh version of this it consists
of layers uh you know one layer might be
um predictions that the story is making
about the future you know that that
could be one uh layer um but uh story
graph is actually more like instead of
sort of static picture like this static
graph a story graph is really more like
a movie it's a evolving graph path

really and and it involves High dimensional vectors of word meaning so there's it's far more complicated than this but this gives you a really rough idea of what I'm after uh cogner has many use cases it if if cogar were successful it would it could be ubiquitous in in in social life um so a group and there again could be all kinds of different groups many different groups Divi uh defines a group event uh its topic its purpose how how decisions were made for this event what kind of methods and rules will be used uh the group that invites people to participate and Communications in this uh for this for this group event could be many to many many to one one to many you know I have examples of all of these in the paper but um and I'll go over a couple uh shortly um use cases could be decision-making customer service emergency reports public polling instructions on how to build a you know a shelf or something how to run a machine system descriptions of how you know how this system works educational uh U use cases and more and the later versions of cogar could include role playing strategizing um you know here's what I believe believe here's what I think this my my competitor believes and I'm going to do this you know how what would my competitor do if I did this or you know all kinds of various versions of Str how do I get the girl should I should I wear

a suit and tie should I wear jeans how
should I act you know like whatever the
whatever the the the issue is at
hand so I this is a example use case for
decision-making that i' I've written out
a little text here but obviously this
again this would be passages that a user
is inputting into the system and um and
a graph is being created you know as
passages are implemented so this person
says you know whatever this issue X is
we better first address issue y that
caused X and issue y started when blank
and it has the effect of blank and if we
don't solve y then we can't solve a and
this is likely to occur that happens and
you know just on and on and on about
some kind of conditional
story complex conditional
story and um in that in this kind of
setting where there where it's a group
decision making what could occur is that
um uh at the top of this craft here uh
story input and editing use there could
be you know 10,000 users or something
10,000 members of this group they each
uh create their story and and uh you
know the the system would be helping
them create the story
too to make it
easy um and then after say the first
round of input the system would would
perform a bunch of computations assess
the what the variety of different
stories are report back on the on how
stories cluster with each other and all
kinds of other things story quality and
you know story components and all kinds

of things so that the group can better understand what the picture of all these stories is and then there would be opportunity for group dialogue feedback story digestion asking questions users asking questions about different other people's uh stories and then after uh that is finished then we can go back to the beginning and uh users see everyone's story with and all the results of of Assessments and then maybe they want to change their story to to reflect what's what has been said maybe I I'm involved in this in this group and I I put down my story and then I eventually get to see what other people are saying and I go and I think oh that's a that's a good idea I want to include I want to include that piece in mind I like that or I don't like this or I'm going to comment on this one of the this part of this guy story so over a series of rounds the idea is hopefully that the belief the belief of the group would start to CES around you know core models you know but the maybe maybe one person story really captures the the group sentiment or maybe maybe the group as a whole the synthesis of their stories captures the group's sentiment and um and over a series of rounds the the thinking of the group the belief model of the group starts to coales around central themes and then there could be voting at the end various kinds of voting that would be possible another use case is emergency

reports again this is a text passage but it would be formed into a story graph but somebody might say hey I might call a utility and say I just I just witnessed an accident uh near this intersection a large truck uh began swerving uh it was wet shouldn't have been going so fast you know uh there have been many accidents at the corner as a kind of an aside uh truck hit an electric pole the wires started Sparks in the grass there's a school nearby uh you know all of this is there's all kinds of implications about this story right the implication is is that that there's some kind of danger maybe the maybe the the operator should send a fire truck a ambulance you know send the police you know something like that there's dangers to the there's implied dangers to the utility workers um the this is you know just a short little paragraph but it's actually quite rich and there's opportunity for many kinds of there's quite a lot of of of information that can be inferred from this little story that isn't said explicitly um in the second paper I list a deera of about 20 uh aspects of what a what a meaning representation should be like or should be capable of doing uh I just list a few here but it should be graph-based to you know to be amenable to computation uh should be capable of handling hand Ling document length stories you know the equivalent of say you know a few pages of text or maybe even a chapter of text

in a book or something like that and
hand and should be able to handle a wide
variety of linguistic phenomena uh
should be re reducible in the sense that
similar stories should make similar
graphs and there's more and more and
more should components should be
compositional um uh and reusable and
more I won't go into any more detail
that but there is a whole list of of uh
requirements so the last couple slides
are just sort of you know fun to think
about how how some of this might be done
and what might happen in the future uh
are there applications of category
Theory here um this last paper on group
interactions belief sharing via sheath
and topus Theory I think is actually uh
uh might be quite appropriate because
that's exact what is happening in this
group setting the the group is sharing
their belief models those belief models
are are are
are sh you know the the the being being
they evolve according to how the group
thinking is going they're they're
unifying generally the group would be
unifying around one or a few
stories and I think all of that could be
studied
mathematically uh using some ideas from
category Theory but also in category
Theory you know one of the ideas of
category theory is to reduce the
complexity of a of a project or you know
system so that's important because this
is obviously a complicated system uh C
set property graphs might be a basis for

story graphs funs might might be allow
for graph to graph graph to text and
other kinds of

Transformations um some ideas from
category Theory might allow us to zoom
in and zoom out this if you can imagine
some very complicated story graph but
the user just wants to focus in on just
one little piece to verify that the
system understood emmer her correctly
and for that you might want to do a kind
of a zoom in or zoom out summarization
of some com some more complicated
components of a story
graph so there's applications for
category

Theory um and
there's I think applications for
friston's um from pixels to planning
paper on renormalizing generative models
um that I think there's potential there
too because uh a graph really represents
a path you know an
evolving path of the graph itself is
evolving right there's it's like a movie
as I said

earlier and uh so the renormalizing
approach might be potentially useful
there um and there's many many
opportunities for agent-based agent
agentic models uh for example group
members understanding each other's
beliefs so that's uh that's about it um
yeah and again the project page has all
kinds of information about
this thank you

John Ian if you have a question
otherwise I can read one from the live

chat oh yes maybe a quick one before the live

chat so it's a you know lovely optimistic um sort of view that every member's story can be heard and understood and contribute to the the collective um and it's sort of making me think about you know CI citizen kind of involvement in cities and knowing what issues are important to a community um then is there a way that it can help with sort of this perennial issue of top down opinion and bottom up needs and wishes um so how would you wait how would you wait let's say the the the stories from um you know opposing members right of of a group if there's yeah right

um so so there's the bigger issue that you know around what you've just said and that's how do you keep this whole thing to actually represent functional cognition like and uh obviously if one party has enormous power and another party has little power that's the you know the the cognitive process is not going to be functional so um it quite clearly a lot of work would have to go into the design of this to to prevent uh uh you know one party from dominating the entire discussion for example and there's and and work would have to go into preventing you know hijacking of the process for some you know dysfunctional means or purpose but uh it you some of that is quite addressable so it should be I I it seems to me that it would be quite possible to create a

system that that is fair uh if everyone has only one voice that's one way to to prevent um one you know one person from Domina in discussion that is if this on a very simple level if there's not Bots you know if a person can't create uh imaginary Bots to be broadcasting their version of what should happen then the person would only have one voice and and over time maybe a group would get to know its members a bit and some members would be more respected than others just naturally because they've contributed in the past they've had intelligent things to say they've you know they they are experts in this field or whatever so um we only have a minute here so I can't I can't go into much further of how that would you know the ways that kind of fairness and transparency and all that kind of stuff would be addressed but I I would think that there's many opportunities to to do so thank you John awesome presentation and looking forward to continuing the collaboration so see you e